
ZOOLOGY

Individual Growth of *Lymnaea stagnalis* (Lymnaeidae, Gastropoda): II. Late Postlarval Ontogeny

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Abstract—Our investigation on the growth of 14 individuals of the great pond snail *Lymnaea stagnalis* was performed in an aquatic culture at 18°C beginning from the 10th week after hatching until death. It has been demonstrated that the increase in the mollusk mass follows an S-like curve during the whole studied period. Linear growth (conch height) follows a parabolic (convex) curve until the age of 39 weeks. Both weight and linear growth during studied period significantly approximate to the Bertalanffy equation, while the interrelation between mass and conch height corresponds to the allometric equation. The meanings of the coefficients of these equations do not differ significantly in different individuals. At the age of 38 to 39 weeks, all mollusks demonstrate breakage in the curve of linear growth, then followed with abrupt slowing of growth until stopping or even decreasing in size in some cases. Neither the Bertalanffy equation nor the allometric relation describe the linear growth of individuals with ages exceeding 39 weeks.

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INTRODUCTION

The postlarval ontogeny of animals can be divided into two periods, which differ in growth character. In the early period (i.e., after final metamorphosis to juvenile individual and transition to the life mode of mature animals), processes can be observed that can be connected with adaptation to new conditions. These processes occur in different way in different species. In this connection, no common rules concerning animals' growth during early ontogeny can be found. Growth curves can vary significantly: concave (Blaxter, 1969; Ivanova, 1973; Cook et al., 2007), convex (Buitrago et al., 2005; Allen, Podolsky, 2007; Amaral et al., 2007; Holst et al., 2007), S-like (Gordon et al., 2004), curve convex on early stages and concave on later stages (Ozernyuk, 1985), and straight line (Blaxter, 1969).

At later stages, growth follows some general principles. Two types of adult growth can be described, viz. unlimited (when size increases over the whole life) and limited (with maximal sized achieved during the life of the animal). The differences between these growth types are as a whole insignificant, as the same equation can be used for their description (Zotin, 2006).

In most works devoted to growth studies, the dependence of mass on age in a group of animals is investigated (Thompson, 1942; Vinberg, 1975; Zotin, Zotina, 1993). However, as was noted by Mina and Klevezal' (1976), the joint (group) data obtained in such studies, on the one hand, seem to "mask" the peculiarities of individual growth and, on the another hand, can lead to artifact results (e.g., in the case of selective choice of individuals with reference to their size).

Investigations on growth of certain individuals are carried out rather rarely (Brody, 1945; Frank, 1969; Phillips, 1981; Cledón et al., 2005; Zotin, Ozernyuk, 2006). Such work needs either individual marking of animals or their cultivation apart from each other. As data obtainment can occupy a long period (especially in long-living species), only a certain ontogeny stage can be evaluated (the period of mass increase is usually chosen).

This complicates the quantitative description of growth with usage of certain equations. In the case of animals with the "unlimited" growth type (hydrobionts mostly), limitations on the investigated period are of less significance and lead only to less precision of the growth parameter evaluation. However, measurements need to proceed in animals with "limited" growth even after the cessation of mass increase, as the mass maximum can be followed by stabilization or decrease (most desirably until the death of the individual).

All the listed facts ensure that species with short life cycles and easy adaptation to conditions of laboratory cultivation are needed for quantitative studies on individual growth.

The given work follows two series of investigations. The first of them is devoted to growth and energy metabolism in *Lymnaea stagnalis* ontogeny. In earlier works, the embryonic (Zotin, Kleimenov, 2006) and early postlarval (first ten weeks after hatching; Zotin, 2009b) stages of ontogeny of this species were studied. The second series is connected with observations on individual growth of different gastropod species. Earlier studies on individual growth of the reticulated snail

Deroceras reticulatum (Zotin, 2007) and field snail *Deroceras agreste* (Zotin, 2009a) were carried out.

MATERIAL AND METHODS

The mollusks *L. stagnalis* L. (Gastropoda, Lymnaeidae) were obtained via cultivation in laboratorial conditions. In total 14 individuals which passed the postlarval period (Zotin, 2009b) were individually cultivated in settled (for more than two days) water from the supply system at constant temperature of 18°C in plastic glasses of 500 ml covered with a nylon net with cells of 0.5 mm. The water and food were replaced twice a week. White cabbage (*Brassica oleraceae* L.) served as food for the mollusks.

The measurements were performed every one or two weeks until natural death of the animals.

The height (maximum size) of the conch was measured with callipers with a maximal precision of 0.1 mm. The total mass of the mollusks' body was measured using torsion scales (precision up to 1 mg) for mass not exceeding 250 mg, or on technical scales (precision up to 10 mg) for larger animals.

The obtained data were approximated using the MatLab software package (7.2.0.232 version) with the following equations.

Allometric dependence:

$$M = aH^k, \quad (1)$$

with M standing for total mass, H for conch height, a and k for coefficients.

Bertalanffy equation (Bertalanffy, 1960):

$$Y = Y_{\infty}(1 - e^{-k_g(t+t_0)})^u, \quad (2)$$

with Y standing for total weight or height of conch, t for age, Y_{∞} , k_g , t_0 , and u for coefficients.

As the error of determination of coefficient u is high enough, its meaning was proposed to be constant (1 in the case of linear growth and 4 in the case of weight growth). The theoretical basis for such meanings of u is presented in an earlier work (Zotin, 2009b).

The comparison of growing curves was performed with Ford–Walford's equation (Ford, 1933; Walford, 1946), i.e., the linear recurrent form of the Bertalanffy equation:

$$Y_{t+2}^{1/u} = cY_t^{1/u} + d, \quad (3)$$

with Y_t standing for the growth parameter (conch weight or height) at the age of t weeks, Y_{t+2} growth parameter at the age of $t + 2$ weeks, u for the constant from Bertalanffy equation (2). The coefficients from equation (3) exhibit the following correlation with ones from Bertalanffy equation (2): $c = \exp(-2k_g)$; $d = Y_{\infty}^{1/u}(1 - c)$.

The estimation of the adequacy of equations (1) and (3) for approximation of experimental data was performed with the criterion of nonlinearity (Zotin, 2000).

Comparison of the coefficients of equations (1) and (3) was performed using methods of regression analysis (Khal'd, 1956).

RESULTS

It was demonstrated in an earlier work (Zotin, 2009b) that mass death of mollusks occurs on early stages of postlarval ontogeny. As a result, only 14 of 517 studied individuals (~2.5%) reached adult age. The age of these surviving individuals' adaptation to a new environment and, hence, of the beginning of late postlarval ontogeny varies from 10 to 14 weeks. The mollusks lived for 55 to 95 weeks reaching a maximal mass of 3.21–6.94 g and maximal conch height of 31.8–40.9 mm.

Data on the dynamics of conch height and total mass in individuals of *L. stagnalis* are presented in the table. Growth curves in different individuals exhibit a similar pattern. Mass increase is observed during the whole ontogeny until death and follows an S-like curve with a single inflection (Fig. 1).

A different pattern is observed for linear growth (Fig. 1). In the age interval between 10 and 39 weeks old, the increase of conch height represents a convex curve lacking inflections. On the 38th or 39th week of life of all individuals studied, a breakpoint in the curve is observed, then, it is followed by an abrupt decrease in the growth rate. The subsequent increase in linear sizes is very slow until complete cessation. In certain cases after reaching the maximum point, the conch height decreases (Fig. 1a, mollusk no. 1). In this connection, the period after the 39th week was designated as the stationary phase of linear growth.

The averaged growth curves for all individuals exhibit insignificant differences from the ones for individual mollusks (Fig. 1d).

The weight dynamics for the whole studied interval of ages and linear growth before breakpoint are approximated well with Bertalanffy's growth equation (2). The comparative analysis with Ford–Walford's (3) equation reveals the absence of statistically significant differences of meanings of coefficients in different individuals.

The average meanings of coefficients are the following: for weight growth, $Y_{\infty} = 4.57 \pm 0.91$ g, $k_g = 0.0645 \pm 0.0027$ weeks⁻¹, $t_0 = -3.1 \pm 0.2$ weeks ($u = 4$, number of measurements $n = 422$); for linear growth, $Y_{\infty} = 39.5 \pm 0.23$ mm, $k_g = 0.0621 \pm 0.0098$ weeks⁻¹, $t_0 = -7.0 \pm 0.2$ weeks ($u = 1$, $n = 125$).

No patterns of linear growth were revealed in the stationary phase. The equation of a straight line seems (Fig. 1) to correspond best to description of the observed changes, although statistical analysis evidences a nonlinear connection between conch height and age with probability $p < 0.001$.

The interrelation between conch mass and height can be approximated significantly with allometric

Changes in conch height (H , mm) and total body mass (M , g) during individual development of *L. stagnalis*

t	No. of mollusks, hatching data													
	1 01.12.2004		2 02.12.2004		3 01.12.2004		4 01.12.2004		5 21.10.2004		6 01.12.2004		7 01.12.2004	
	H	M	H	M	H	M	H	M	H	M	H	M	H	M
10	—	—	4.6	0.01	7.5	0.03	—	—	9.2	0.06	—	—	9.7	0.05
12	—	—	13.6	0.11	11.2	0.09	14.6	0.14	12.6	0.11	—	—	12.6	0.12
14	12.2	0.11	16.2	0.36	15.4	0.22	17.5	0.32	16.2	0.37	12.6	0.20	16.0	0.19
16	15.1	0.25	19.4	0.57	16.5	0.34	20.0	0.48	18.6	0.59	13.6	0.22	17.0	0.38
18	16.5	0.35	21.1	1.03	20.6	0.52	24.5	0.80	19.5	0.59	17.4	0.32	24.3	0.64
20	19.9	0.45	26.1	1.02	24.8	0.84	25.3	0.86	25.0	0.90	18.9	0.45	24.4	0.74
22	21.1	0.59	26.4	1.17	25.4	1.03	26.3	1.01	25.8	0.91	20.5	0.51	26.3	0.92
24	26.9	1.24	26.4	1.80	27.8	1.58	28.8	1.42	27.9	1.58	23.3	0.77	29.0	1.31
26	27.7	1.46	32.0	2.61	28.7	1.73	30.3	1.90	28.3	1.61	25.0	1.13	29.8	1.31
29	29.9	1.92	33.8	3.21	31.6	2.40	32.9	2.38	30.0	1.99	26.4	1.28	30.5	1.51
31	31.6	2.25	35.5	3.43	32.7	2.52	33.5	2.80	31.3	2.22	27.4	1.58	32.0	1.78
33	33.0	2.56	36.1	3.82	32.8	2.84	34.7	3.05	31.3	2.50	27.2	1.68	32.2	1.76
34	33.4	2.89	36.3	4.05	33.0	2.74	34.5	3.38	31.9	2.56	27.5	1.78	33.0	2.19
38	36.4	3.75	36.4	4.10	34.6	3.17	34.6	3.42	34.0	2.24	30.8	2.50	36.2	2.90
39	36.4	4.11	37.2	4.60	35.4	4.10	35.2	3.55	34.5	3.30	30.2	2.72	36.5	3.06
41	36.3	3.73	37.3	4.31	35.5	3.78	35.3	3.61	34.9	3.46	31.3	2.49	35.6	2.92
43	36.4	3.69	37.4	4.65	35.4	3.96	35.2	3.19	34.7	3.45	30.8	2.57	36.3	2.99
45	36.1	3.67	37.4	4.68	35.4	4.15	35.3	3.25	34.7	3.57	30.6	2.67	36.3	2.93
47	36.3	3.96	37.4	4.57	35.7	4.00	35.4	3.56	34.9	3.57	30.8	2.51	36.4	2.95
49	36.7	4.09	37.6	4.66	36.1	4.32	35.4	3.54	34.8	3.52	31.0	2.71	36.6	3.19
51	36.6	4.07	38.3	4.91	36.3	4.51	35.4	3.72	34.8	3.51	31.3	2.62	36.5	3.10
52	36.6	4.14	38.5	5.19	35.9	4.64	35.4	3.72	35.1	3.67	31.3	2.84	36.6	3.16
53	36.6	4.16	38.3	5.07	36.6	4.56	35.4	3.68	35.0	3.62	31.3	2.79	36.6	3.11
54	36.8	4.13	38.4	5.04	36.6	4.70	35.5	3.80	35.0	3.64	31.5	2.80	36.8	3.24
55	36.8	4.06	38.8	5.17	36.6	4.81	35.7	3.61	35.0	3.63	31.2	2.68	37.0	3.09
56	36.9	4.24	38.4	5.14	36.9	4.72	35.4	3.72	35.1	3.78	31.8	2.92	37.0	3.28
57	37.0	4.14	38.4	5.08	37.0	4.82	35.5	3.81	35.2	3.70	31.3	2.78	37.0	3.26
58	36.9	4.19	38.7	5.36	37.3	5.20	35.4	3.89	34.9	3.67	31.4	2.94	36.8	3.28
59	36.8	4.29	38.2	5.26	37.2	5.20	35.5	3.73	35.2	3.71	31.5	3.00	36.9	3.26
60	36.8	4.21	38.4	5.27	37.2	5.17	35.3	3.90	35.1	3.76	31.4	2.90	36.8	3.26
61	36.8	4.14	38.5	5.14	37.2	5.07	35.6	3.68	34.9	3.65	31.4	2.88	36.9	3.16
63	37.1	4.30	38.7	5.16	37.4	5.07	35.7	3.81	35.3	4.00	31.6	2.85	36.9	3.24
65	37.2	4.26	38.7	5.34	37.4	5.31	35.7	3.83	35.2	4.18	31.8	3.03	37.0	3.44
66	37.1	4.28	38.3	5.44	37.5	5.45	35.6	3.97	35.5	4.32	31.8	2.96	37.3	3.44
68	37.2	4.33	38.7	5.25	37.7	5.20	35.8	3.86	35.5	4.29	32.0	3.08	37.4	3.43
69	37.2	4.25	38.6	5.35	37.7	5.39	36.0	3.97	35.4	4.19	32.1	3.29	37.2	3.47
70	37.0	4.34	38.5	5.60	37.7	5.55	36.2	4.10	35.5	4.13	31.9	3.01	37.2	3.54
71	36.9	4.32	38.7	5.67	37.7	5.65	36.1	3.93	35.6	4.20	31.9	3.20	37.2	3.42
72	36.5	4.44	38.8	5.25	37.8	5.38	35.5	4.08	35.7	4.07	31.9	3.03	37.2	3.34
73	36.6	4.45	38.8	5.65	37.8	6.04	36.2	4.15	35.7	4.24	31.9	3.07	37.2	3.72
74	36.8	4.18	38.8	5.40	38.2	5.27	35.9	4.14	35.5	4.14	31.8	2.94	37.3	3.50
75	36.6	4.40	38.7	5.23	38.2	6.01	36.0	4.39	35.4	4.40	31.6	3.44	37.1	3.92
76	36.8	4.36	38.4	4.94	37.4	5.93	36.1	4.20	35.4	4.40				
77	36.6	4.32	38.4	5.62	37.2	6.00	36.3	4.90						
78	36.6	4.47	38.8	5.62	37.4	5.73	36.3	4.13						
79	36.6	4.29	38.8	5.61	37.3	5.80	36.1	4.26						
80	36.4	4.87	39.4	6.15	37.5	6.94	36.4	4.01						
81	36.4	4.46	38.9	6.04	37.4	6.16								
82	36.2	4.51	39.2	5.66	38.1	6.83								
83	36.4	4.82	39.4	5.91										
84	36.4	4.20	39.4	5.57										
85	36.3	4.65	39.2	6.19										
87	36.3	4.39	39.2	6.06										
91	36.3	4.40	39.2	6.10										
92	36.1	4.20												
93	35.8	4.06												
94	36.1	4.43												
95	36.0	4.36												

Table. (Contd.)

<i>t</i>	No. of mollusks, hatching data													
	8 03.12.2004		9 26.11.2004		10 01.12.2004		11 02.12.2004		12 03.12.2004		13 01.12.2004		14 05.12.2004	
	<i>H</i>	<i>M</i>	<i>H</i>	<i>M</i>	<i>H</i>	<i>M</i>	<i>H</i>	<i>M</i>	<i>H</i>	<i>M</i>	<i>H</i>	<i>M</i>	<i>H</i>	<i>M</i>
10	–	–	6.8	0.02	–	–	6.8	0.02	–	–	11.7	0.09	–	–
12	5.6	0.01	11.7	0.08	–	–	14.1	0.13	4.6	0.01	16.5	0.29	–	–
14	9.2	0.06	16.5	0.19	–	–	17.8	0.40	9.7	0.04	18.6	0.49	4.9	0.01
16	11.2	0.09	17.8	0.37	8.5	0.04	18.6	0.51	12.6	0.07	21.6	0.77	6.1	0.01
18	12.6	0.11	19.4	0.43	10.7	0.07	19.4	0.64	13.6	0.19	22.4	0.86	10.7	0.05
20	14.3	0.18	23.5	0.56	16.7	0.22	26.4	1.03	19.6	0.37	23.8	1.03	14.6	0.20
22	14.2	0.18	24.4	0.70	17.9	0.24	25.4	1.13	22.5	0.56	24.0	1.13	17.0	0.37
24	14.9	0.19	24.9	0.86	17.8	0.30	29.2	1.77	28.3	1.25	26.9	1.54	17.8	0.52
26	16.6	0.38	28.8	1.69	22.5	0.62	31.9	2.70	31.4	1.82	27.8	1.78	22.5	0.60
29	21.8	0.72	33.0	2.56	26.9	1.19	34.0	3.40	34.5	2.58	29.1	1.98	22.9	0.51
31	25.5	1.11	34.4	2.77	28.7	1.41	35.3	3.54	35.5	3.03	29.8	2.12	23.2	0.65
33	26.3	1.31	34.7	3.38	28.5	1.63	35.7	3.92	37.4	3.65	29.4	2.30	26.3	1.07
34	27.2	1.53	35.2	3.20	29.9	1.98	35.9	3.74	38.3	3.93	29.6	2.33	28.4	1.33
38	29.7	2.00	35.5	3.29	30.4	2.09	36.8	4.48	39.1	4.17	31.2	2.94	32.2	1.96
39	31.6	2.69	36.6	4.02	33.3	2.83	36.8	4.40	39.7	4.51	31.3	2.97	35.0	2.59
41	32.0	2.70	36.3	3.96	33.8	2.74	36.7	4.36	39.9	4.56	31.2	2.86	35.3	2.84
43	32.7	2.68	36.7	3.85	34.0	2.95	36.8	4.25	40.0	4.25	31.6	2.88	35.5	2.79
45	32.7	2.86	37.1	4.03	34.2	3.03	36.9	4.47	40.0	4.55	31.6	2.81	36.0	3.46
47	32.4	2.90	37.4	4.28	34.3	3.03	36.8	4.62	40.4	4.33	31.6	2.77	36.8	3.65
49	32.9	3.22	38.0	5.14	34.9	3.50	37.3	4.97	40.6	4.70	31.8	2.95	37.0	3.61
51	33.4	3.39	37.9	4.56	35.2	3.69	37.4	4.26	40.6	4.44	31.6	3.00	36.8	3.83
52	33.7	3.32	38.2	4.75	35.2	3.70	37.4	5.28	40.8	4.78	31.7	3.12	–	–
53	33.3	3.41	38.4	4.88	34.6	3.69	37.7	4.73	40.9	4.52	31.8	3.12	37.0	3.95
54	33.8	3.36	38.5	4.91	34.9	4.10	37.4	5.11	40.5	4.63	–	–	–	–
55	33.6	3.28	38.2	4.65	35.2	4.10	37.3	4.75	40.5	4.19	31.8	3.21	37.3	3.85
56	33.8	3.38	38.8	4.80	34.5	4.52	36.9	5.41	40.5	4.43			–	–
57	33.7	3.25	38.9	5.05	35.2	3.89	37.4	4.90	40.4	3.91			37.4	3.98
58	33.8	3.33	39.3	5.09	35.2	4.24	37.4	5.25	40.4	4.52			–	–
59	33.8	3.36	39.1	4.90	35.2	4.14							37.7	4.37
60	33.7	3.46	39.0	5.11	35.3	4.23								
61	33.8	3.38	39.4	5.21	35.5	4.15								
63	34.1	3.38	39.7	5.07	35.5	4.27								
65	33.8	3.39	40.0	5.38	35.6	4.21								
66	34.0	3.59	40.0	5.67	36.0	4.56								
68	34.0	3.42												
69	33.9	3.47												

Note: *t*, age after hatching (weeks); dash, measurements not performed; empty cell, mollusk dead by this date.

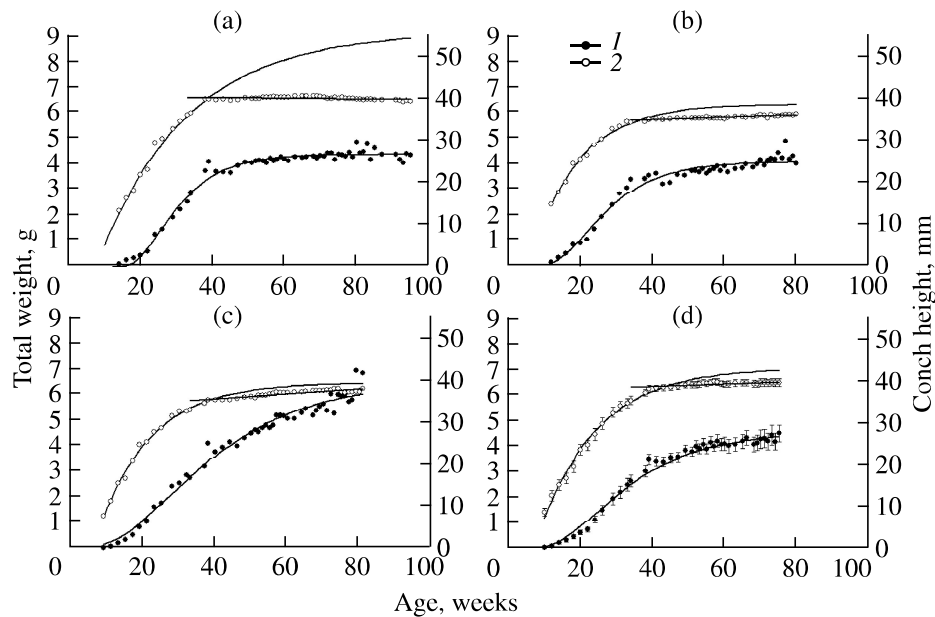


Fig. 1. Changes in mass (1) and conch height (2) of *L. stagnalis* in dependence on the age after hatching. (a–c), individuals nos. 1, 4, and 5, respectively; (d), averaged meanings for all individuals. Curves represent approximations with Bertalanffy's equation (2), and straight lines indicate tendency of conch height changes in stationary phase of linear growth.

equation (1) (Fig. 2). The coefficients of this connection lack a statistically significant difference and present the following meanings: $a = 0.0411 \pm 0.004 \text{ mg/mm}^k$, $k = 3.16 \pm 0.03$ ($n = 185$).

In the stationary phase, the interrelation between the logarithms of total weight and conch height is significantly nonlinear ($p < 0.001$) and, hence, equation (1) is eligible for description of the connection between these parameters.

DISCUSSION

The obtained results evidence that weight growth of *L. stagnalis*, as compared with that of ground gastropods (Zotin, 2007, 2009a), has unlimited character and can be described by Bertalanffy's equation (2). This type of growth is typical for most hydrobionts (Ivanova, 1973; Vinberg, 1975; Mina, Klevezal', 1976; Alimov, 1981).

The presence of recurrent linear form (3) of Bertalanffy's equation gives the opportunity for comparative analysis of growth curves with methods of linear regression analysis. The results of such analysis indicate an absence of significant differences in growth parameters in different individuals. This means that the observed variations in the mass limit of mollusks (Y_∞) and rate of its attainment (k_g) in different individuals can be interpreted as random.

The same can be concluded for linear growth in the age interval between 10 and 39 weeks. However, later stages demonstrate deviations of growth curves from the ones graphed according to Bertalanffy's equation (Fig. 2). Such deviations occur in all studied mollusks

at about the same age and represent an abrupt breakage in the curve, after which the increase in the conch height is almost completely finished. The same pattern of linear growth was demonstrated in some sea gastropods (Darling, 1965; Frank, 1969).

Some data on linear growth cessation are present for other animal species. The most well-known example is humans, which cease growth at the age of 17 to 22 (Harrison et al., 1979).

At first glance, the limited type of growth can be proposed for both these species and *L. stagnalis*. In this case gastropods represent an exception from the common rule of the unlimited growth of hydrobionts (Vin-

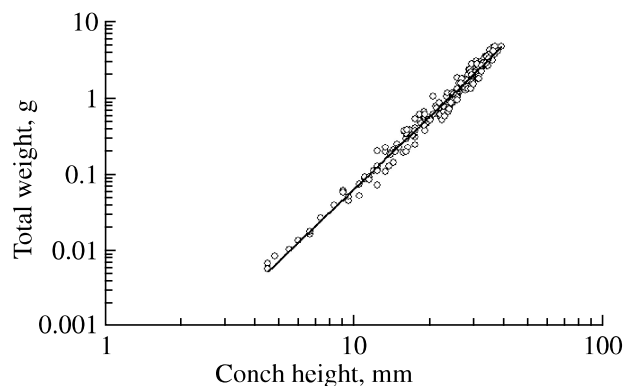


Fig. 2. Dependence of total weight on shell size in *L. stagnalis* in the age interval between the 10th and 39th weeks after hatching. The straight line indicates approximation with allometric equation (1). Logarithmic scale.

berg, 1975; Mina, Klevezal', 1976; Alimov, 1981). However, the limited growth type represents gradual reaching of the maximal sizes and their subsequent decrease, which can be described with equation based on formulas of thermodynamics of irreversible processes (Zotin, 2006) and probably needs no additional regulation from the organism.

The abrupt change in the growth curve gives the opportunity to propose the existence of such regulation, which can be connected with puberty of mollusks and, hence, with the necessity of redistribution of energy expenditure from growth to reproduction processes.

The connection between growth cessation and puberty was also noted in some species of terrestrial gastropods (Abeloos, 1944; Hunter, 1968; Dmitrieva, 1969; Likharev, Viktor, 1980; South, 1982; Duval, Banville, 1989; Barker, 1991).

Allometric pattern (1) usually used for description of the interrelation between mass and conch height (Alimov, Golikov, 1974) can be applied only during the period of conch height increase (up to 39 weeks after hatching). The power coefficient of this dependence $k = 3.16 \pm 0.03$ does not conform to $k = 2.79 \pm 0.20$ from the work of Alimov and Golikov (1974) calculated for family Lymnaeidae. This can be connected with fact that cited authors took into account the entire range of mass and conch height meanings, including ones for early postlarval ontogeny and the stationary phase of linear growth. However, in early postlarval ontogeny the coefficient $k = 2.33 \pm 0.03$ significantly ($p < 0.001$) differs from the one in late ontogeny (3.16). On the another hand, allometric equation (1) cannot be applied to stationary phase (age after 39 weeks) for description of the correlation between mass and conch height in *L. stagnalis*.

If we combine the data obtained in this and earlier investigations (Zotin, Kleymenov, 2006; Zotin, 2009b), the total pattern of growth of *L. stagnalis* during ontogeny includes the following stages.

Weight growth:

(1) Embryonic development (3–0 weeks before hatching), S-like growth type (weight range from 1.4 to 400 µg).

(2) Early poslarval ontogeny (0–10 weeks after hatching), linear or weakly exponential growth type (weight range from 0.4 to 3 mg).

(3) Late postlarval ontogeny (from ten weeks until death), S-like unlimited growth type (weight range from 0.003 to 3–7 g).

Linear growth (embryo length or conch height):

(1) Embryonic development (3–0 weeks before hatching), S-like growth type (length range from 0.14 to 1.25 mm).

(2) Early poslarval ontogeny (0–10 weeks after hatching), linear or weakly exponential growth type (length range from 1.25 to 3 mm).

(3) Late postlarval ontogeny, growth phase (10–39 weeks after hatching), parabolic unlimited growth type (length range from 3 to 31–40 mm).

(4) Late postlarval ontogeny, stationary phase (from 39 weeks after hatching until death), insignificant alterations of sizes, both increase and decrease (variations of individuals 30–40 mm).

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